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Diogo B. Provete¹, Lilian Franco-Belussi², Lia R. Santos⁵, Rodrigo Zieri², Rafaela M. Moresco², Itamar A. Martins⁴, Silvio C. Almeida³, Classius Oliveira²

¹*Universidade Federal de Goiás, Goiânia, GO, Brazil*, ²*Universidade Estadual Paulista, São José do Rio Preto, SP, Brazil*, ³*Universidade Estadual Paulista, Botucatu, SP, Brazil*, ⁴*Universidade de Taubaté, Taubaté, SP, Brazil*, ⁵*Instituto Federal Goiano, Rio Verde, GO, Brazil*

Phylogenetic signal and variation of visceral pigmentation in eight anuran families

Visceral pigmentation is found in several organs and structures of ectothermic animals, comprising the extracutaneous pigmentary system. Its function is not well defined, although it is known that melanin is produced and stored inside pigmented cells. Previous studies demonstrated that the distribution of visceral pigmentation is neither homogeneous among organs nor among anuran species. We describe the diversity of visceral pigmentation in 12 organs/structures from 32 anuran species belonging to eight families using a comparative phylogenetic method. We also determined in which node(s) of the phylogeny there is more variation in the pigmentation categories and whether this variation has phylogenetic signal. We tested for the pigmentation in all organs and each organ separately. The visceral pigment cells in organs and structures of the abdominal cavity varied among genera. All species had pigmentation in the urogenital and cardiorespiratory systems, whereas the stomach lacks pigmentation in all species. We also found a phylogenetic signal for pigmentation in all organs and structures taken together, besides heart, testes, lumbar parietal peritoneum and lumbar nerve plexus when considered separately. Overall, considering all organs, the highest diversity of categories of pigmentation was found in the nodes corresponding to Craciabatrachia and Athesphatanura. This study constitutes the first step towards understanding the evolution of visceral pigmentation in anurans.